

INDUSTRIAL MACHINE FAILURE PREDICTION

Track 1 Final Capstone • Binary Classification • Lightweight ML Prototype

1. Abstract

This project develops a lightweight binary-classification prototype that estimates whether an industrial machine is likely to experience a failure within a defined operating window. The workflow covers data quality, exploratory analysis, preprocessing, foundational model comparison, evaluation and a Streamlit demonstration.

2. Problem Definition

Predict machine failure (0/1) from temperature, vibration, pressure, runtime and maintenance-gap indicators. The solution is intended as a zero-cost prototype rather than a production or safety-critical system.

3. Dataset Note

The supplied project brief specifies the expected inputs but does not contain the assigned dataset. Therefore this emergency prototype includes a reproducible synthetic dataset with 5,000 observations. It is explicitly a demo dataset and must not be represented as the internship's assigned dataset. If an assigned CSV is available, replace the demo CSV and rerun the pipeline before final submission.

Rows: 5,000 | Failure rate: 6.56% | Train/Test: 80%/20% | Random seed: 42

4. Methodology

Median imputation and standardization are used for Logistic Regression. Class weighting handles the imbalanced target. Logistic Regression and Decision Tree are compared. Logistic Regression is selected for its interpretability, lightweight implementation and probability output.

5. Evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7840	0.2000	0.7576	0.3165	0.8637
Decision Tree	0.9150	0.3431	0.4394	0.3862	0.7785

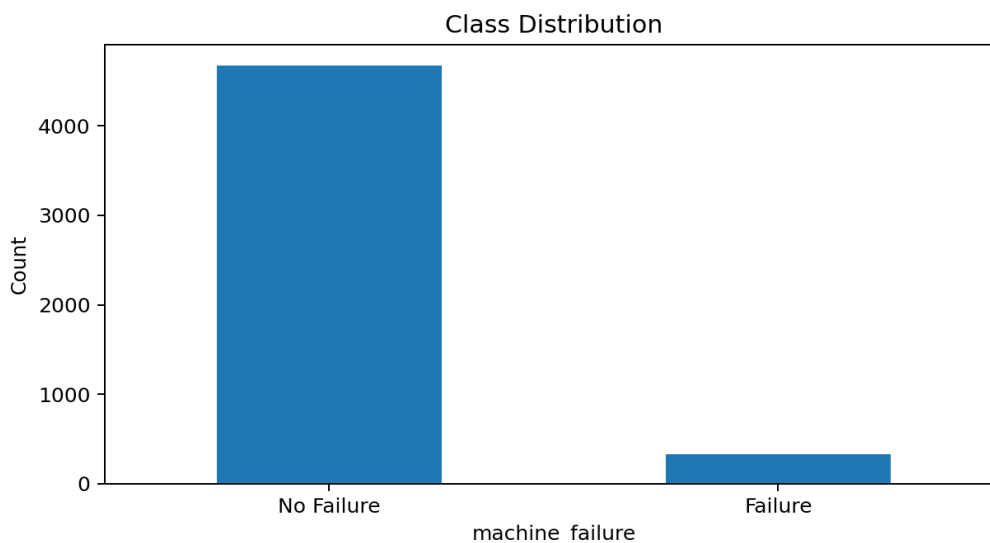


Figure 1. Class distribution.

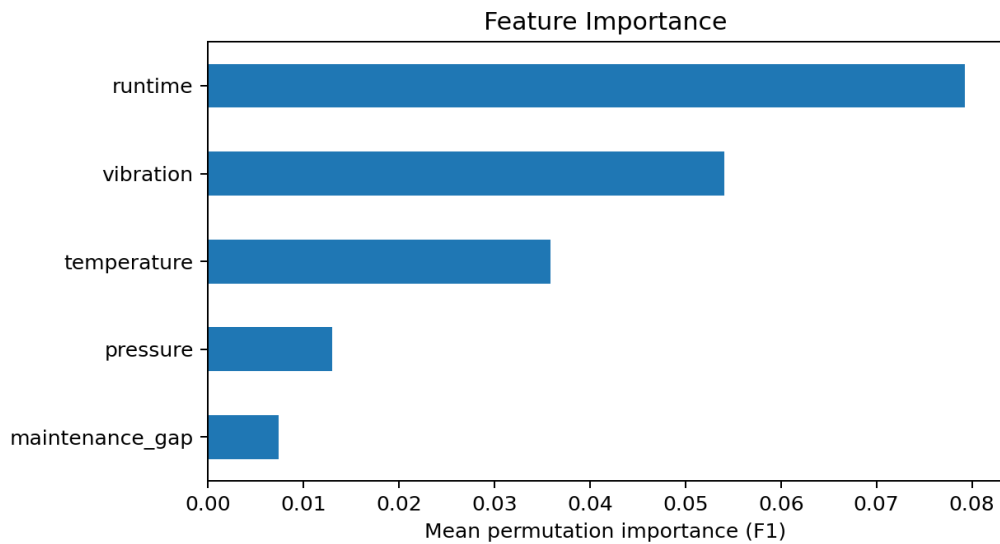


Figure 2. Feature importance.

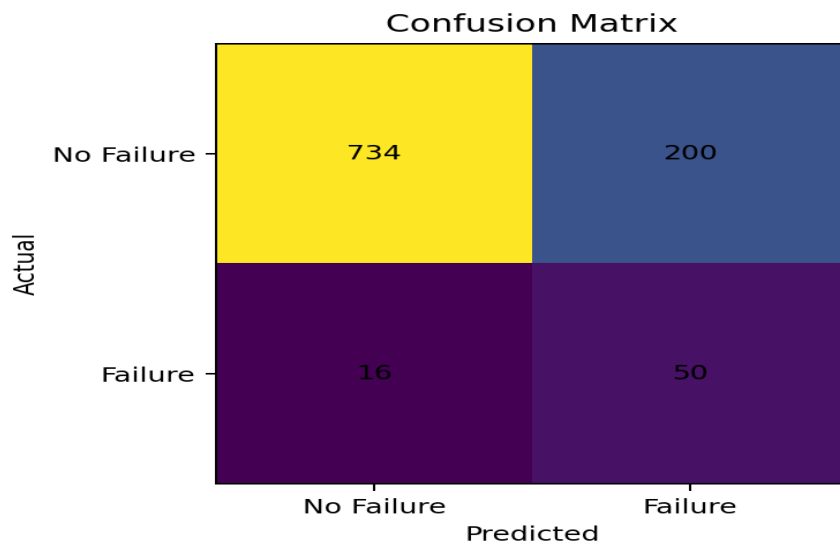


Figure 3. Confusion matrix.

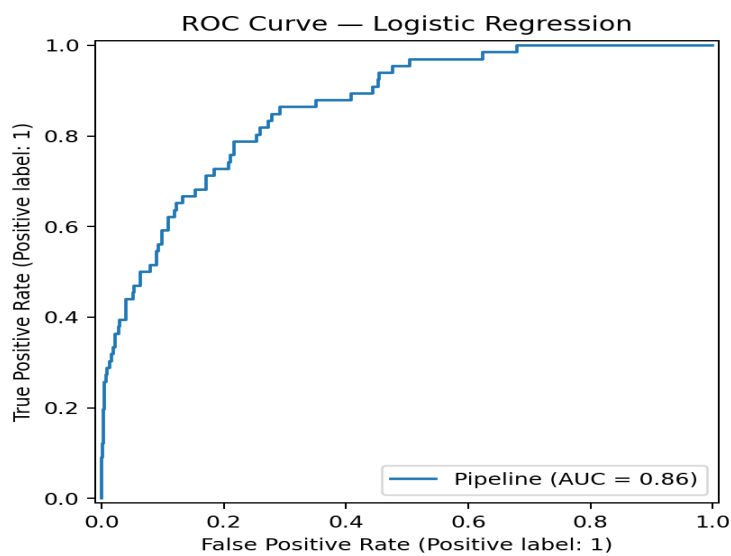


Figure 4. ROC curve.

6. Prototype

The Streamlit app accepts the five machine inputs, applies the same trained pipeline, calculates failure probability and displays a binary risk decision. The interface is intentionally simple for a lightweight demonstration.

7. Limitations

The numerical results are prototype results from the synthetic demo dataset. They should not be interpreted as real industrial performance. The model does not address temporal drift, machine-specific baselines, sensor calibration or safety-critical deployment.

8. Future Scope

Replace the demo CSV with the assigned dataset; add rolling/time-series features; compare additional foundational models; tune the probability threshold using maintenance costs; and add monitoring/drift detection before any real deployment.

9. References

Learn Depth Academy LLP, Track 1 Final Capstone — Problem 21. Scikit-learn documentation: <https://scikit-learn.org/>. If UCI AI4I is used, cite the UCI repository and DOI 10.24432/C5HS5C.